

Sustainable Tourism Management in Juneau

Summary

Juneau, Alaska has become a popular tourist destination in recent years. While tourism spurs economic growth, it also brings significant pressure on environment and local infrastructure. To address this issue, we develop a **multi-objective dynamic programming model** that optimizes **economic**, **environmental**, and **social** outcomes for sustainable tourism management.

For **Problem 1**, we begin with a **total tourist demand model**, considering factors of natural growth, seasonal variations, taxation effect, and government promotion. This model serves as an input to the three interconnected models. First, the **economic model** estimates profit by deducting operational costs from tourism revenue, both of which are assumed to scale linearly with tourist volume. Second, the **environmental model** quantifies tourism's impact through an index aggregating key ecological factors of temperature, ocean temperature, snowfall, and ocean pH using **entropy weight method**. This environment model also accounts for natural recovery and government intervention. Third, the **social impact model** evaluates resident satisfaction based on their attitudes towards public service stress, environmental quality decrease, and tourism taxation.

The objective of our model is to maximize **economic profit**, **environmental sustainability**, and **resident satisfaction** under constraints of **tourist capacity**, **environmental protection measures**, and **government budget limitations**. We use dynamic programming to solve the model and stimulate the results over a 5-year period. Our model recommends a dynamic tax rate in a range of 3.2% to 5.0%, and a schema of government expenditures with large portions of environmental protection and infrastructure construction. Our results predict a stable tourist volume of around **1.6 million** visitors per year. Both government revenue and environment quality rise steadily after implementing our recommended policies. Additionally, we perform a **sensitivity analysis** to identify the most influential factors and assess the robustness of proposed policy measures.

For **Problem 2**, we further validate our model by adapting it to the tourism situation of **Miami**, another destination troubled by **overtourism**. Differences in culture, visitor behavior, and government intervention require adaptation of parameters of our model.

For **Problem 3**, we prepare a memo for the tourism council of Juneau. Based on the predicted results given by our model, we summarize key findings and offer practical recommendations to achieve **sustainable tourism development**.

In conclusion, our multi-objective dynamic programming model provides a comprehensive framework for balancing economic, environmental, and social aspects of Juneau tourism. Our model has certain limitations to some extent, including potential uncertainties in data and simplifications made towards complex human-environment interactions. In future work, we could optimize our model using more granular data and refine our assumptions to further improve the model's predictive capabilities.

Keywords: Multi-Objective Model, Dynamic Programming, Sustainable Tourism